EYLON CAPLAN

Email ◇ Website ◇ GitHub ◇ Google Scholar ◇ LinkedIn

4th-year PhD student in NLP | LLM Evaluation, Retrieval & Context Selection, and Data at Scale

Available for internships Spring–Summer–Fall 2027

EDUCATION

Purdue University, West Lafayette, Indiana, USA — Expected December 2027
Ph.D. in Natural Language Processing, Department of Computer Science — GPA: 3.8/4.0

University of Nebraska–Lincoln, Lincoln, Nebraska, USA — May 2023
B.S. in Computer Science and Mathematics (Minors: Physics, Spanish) — GPA: 3.99/4.0

RESEARCH EXPERIENCE

AWS AI Labs — Summer 2026
Applied Scientist Intern — *Manager: Rashmi Gangadharaiah*

- Built BOUNDARYMORPH, a **Gaussian Process** selector that spends a limited reranking budget choosing which documents enter an LLM's context for **deep-research** queries
- State-of-the-art set retrieval (+5.4 nCG@100); beats a multi-hop **LLM search agent** at 34× lower latency; first-author, under review at ICLR 2027

Purdue University — Aug 2023 – Present
Graduate Research Assistant — *Advisor: Dan Goldwasser*

- **RL** for **open-ended generation** (current): countering **mode collapse** in **GRPO** with real human text (500+ argument strategies mined from 338K argument corpus); early runs win 56.5% on held-out conversation generation
- **Data curation at scale** (SPLITS!): 9.7M-post Reddit corpus split by demographic × topic; triages 23K LLM-generated hypotheses with 15–18× less human review (ACL 2026 Oral, **SAC Highlight Award**)
- **Evaluation** (VIBE): co-designed a **VLM benchmark** isolating a new failure mode, misreading correctly seen social cues; humans beat the best VLM by 17 points (EMNLP 2025 Findings)
- **LLM-in-the-loop retrieval** (CONCEPTCARVE): an LLM iteratively steers a retriever over 165M posts on a fixed budget; +26% over LLM query expansion (ACL 2025)

SKILLS

Research: LLM evaluation, VLM evaluation, benchmarks, retrieval, RAG, reranking, agentic search, context selection, large-scale data curation, synthetic data, post-training, SFT, DPO, GRPO, Gaussian Processes, Bayesian optimization
Tools: Python, PyTorch, HuggingFace (Transformers, TRL, PEFT), vLLM, FAISS, ColBERT, pyserini, SLURM, Docker, Kubernetes, AWS, Dask, Weights & Biases, Hydra, Git

PUBLICATIONS

BoundaryMORPH: Budgeted Reranking via Active Set Selection for Diffuse Retrieval
Eylon Caplan, Shamik Roy, Shib Sankar Dasgupta, Yingfan Wang, and Rashmi Gangadharaiah
Under review at ICLR 2027

Datasets and Methods for Improving the Cultural Capabilities of NLP Systems: A Survey
Tania Chakraborty*, **Eylon Caplan***, Zhaoqing Wu*, Kevin Cushing, Bruce Qin, Shreya Havaldar, and Dan Goldwasser
NLP+CSS 2026, *Equal contribution

TAIGR: Towards Modeling Influencer Content on Social Media via Structured, Pragmatic Inference
Nishanth Sridhar Nakshatri, **Eylon Caplan**, Rajkumar Pujari, and Dan Goldwasser
ACL 2026 Main, Oral

SPLITS! *Flexible Sociocultural Linguistic Investigation at Scale*
Eylon Caplan, Tania Chakraborty, and Dan Goldwasser
ACL 2026 Main, Oral, **SAC Highlight Award (top 0.5%)**

VIBE: Can a VLM Read the Room?
Tania Chakraborty, **Eylon Caplan**, and Dan Goldwasser
EMNLP 2025 Findings

CONCEPTCARVE*: Dynamic Realization of Evidence*
Eylon Caplan and Dan Goldwasser
ACL 2025 Main